

Six weak triple-minimum permutations support at most six labels

Research note: a proof independent of constraint solvers

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All permutation families below are multisets of six equally weighted permutations. A weak triple-minimum family has each member of each triple earliest in exactly two rows. A fully triple-uniform family has every ordering of every triple in exactly one row.

Lemma 1. *No fully triple-uniform six-row family exists on five labels.*

Proof. For a fixed label let d_j be its number of occurrences in position $j \in \{0, 1, 2, 3, 4\}$. Counting labels before it, and pairs of labels before it, gives

$$\sum_j d_j = 6, \quad \sum_j j d_j = 12, \quad \sum_j j(j-1) d_j = 24.$$

The last identity modulo three implies $3 \mid d_2$. The possibility $d_2 = 6$ contradicts $\sum j^2 d_j = 36$. If $d_2 = 3$, put $(a, b, c, d) = (d_0, d_1, d_3, d_4)$. The remaining equations are

$$a + b + c + d = 3, \quad b + 3c + 4d = 6, \quad b + 9c + 16d = 24.$$

Thus $c + 2d = 3$ and $b = 2d - 3$. Since $c \geq 0$ forces $d \leq 1$, this makes $b < 0$. Therefore $d_2 = 0$ for every label, impossible because every row has a label in position two. $\square$

Lemma 1 is the known nonexistence of PSCA(5, 3, 1); the position-moment proof is also given in Gentle and Wanless [1].

Theorem 1. *A weak triple-minimum family of six permutations has at most six labels. The bound is attained.*

Proof. It suffices to exclude seven labels, since deleting labels preserves the property. For distinct labels i, j let r_{ij} count rows in which i precedes j . Triple-minimum uniformity implies $2 \leq r_{ij} \leq 4$.

Let B be the labels i for which $r_{ij} = 2$ for some j , and let A be the other labels. If $i \in B$, the two rows in which $i < j$ must also have $i < k$ for every third label k , because the latter joint event already occurs twice. Thus i is globally first in exactly two rows. There can be at most three such labels, so $|A| \geq 4$.

For any two labels of A , neither order count can be two, hence both are three. On a triple, equal minimum counts together with balanced pair counts force all six orders to occur once. Consequently the restriction to A is fully triple-uniform. Lemma 1 gives $|A| \leq 4$, so $|A| = 4$ and $|B| = 3$. Each member of B is first in two rows, accounting for all six first positions.

Fix $a \in A$. Sum its minimum counts over the three triples $\{a, b, c\}$ with $b, c \in B$. The sum is six. In any row at least one member of B precedes a , so that row contributes at most one to this sum. It follows that every row contributes one: exactly one member of B precedes a . Thus in each row all four labels in A occur between the first member of B and the other two members.

Fix $b \in B$ and distinct $a, c \in A$. Outside the two rows beginning with b , both a, c precede b . The minimum condition on $\{a, c, b\}$ therefore says that $a < c$ holds in exactly two of those four rows. Since it holds in three rows overall, it holds in exactly one of the two rows beginning with b . This holds for every pair from A , so the restrictions of those two rows to A are reverses. The fully triple-uniform family on A is therefore three reversal pairs.

Choose one representative of each reversal pair. For every triple of A , every label must be its middle label in exactly one of these three representatives. A fixed label of A would consequently be middle in three triples in total. But in any permutation on four labels, a label at position $j \in \{0, 1, 2, 3\}$ is middle in exactly $j(3 - j) \in \{0, 2\}$ triples. The sum over three representatives is even, a contradiction.

Sharpness is witnessed by the six permutations

012345, 521340, 531204, 413250, 423105, 032145,

whose triple-minimum counts are checked in `verify_nine_dice_cloning.py`. □

This theorem concerns six equally weighted rows. It does not exclude seven labels with six rows of unequal positive weights, nor does it exclude ten twelve-face three-wise fair dice by another construction. No claim of literature novelty is made here.

References

- [1] B. Gentle and I. M. Wanless, *On Perfect Sequence Covering Arrays*, Annals of Combinatorics (2023). arXiv:2202.01960, doi:10.1007/s00026-022-00610-6.